Engine Model(s): PT6A

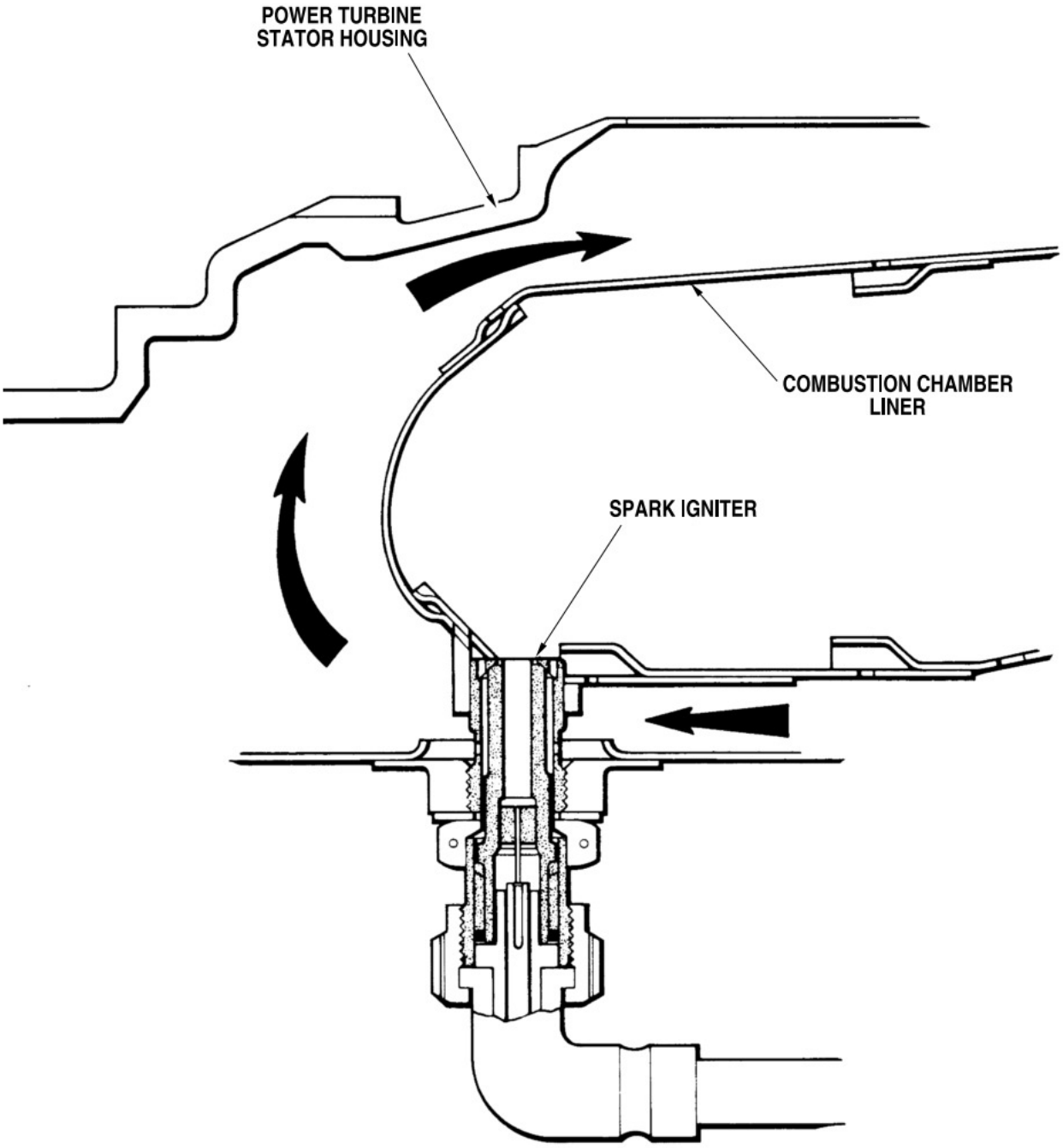
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IGNITION CABLES AND SPARK IGNITERS - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Spark Igniter Installation (Typical) - Cross Section



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The two individual ignition cable assemblies carry the electrical energy output from the ignition exciter to the engine mounted spark igniters. Each lead assembly consists of an electrical lead contained in a flexible metal braiding. Coupling nuts at each end of the assembly facilitate connection to the respective connectors on the ignition exciter and spark igniter. Mounting flanges for attachment to the engine fireseal mount rings are brazed onto the flexible braiding.

The spark igniters, located at the 4 and 9 o'clock positions on the gas generator case and adjacent to the fuel manifold, are in the form of a double-ended, threaded plug with a central positive electrode enclosed in an annular semi-conducting material. The electrical potential developed by the ignition exciter is applied across the gap between the central conductor and the igniter shell (ground). As this potential increases, a small current passes across the semi-conducting material. This current increases until the air between the central conductor and the shell ionizes. When ionization occurs, high energy discharges between the electrodes. The spark always occurs somewhere in the annular space between the central conductor and shell.

Spark igniters are available under several different part numbers, to cover parts obtained from different sources. Although parts may thus differ in detail appearance, the functioning is identical and they are fully interchangeable. The latest configurations feature air cooling holes for greater durability.

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